

INFSCI - 2591 Algorithm Design

Homework-3

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1. Memoization approach code:

```
def howSum_memo(targetSum, numbers, memo=None):
    if memo is None:
        memo = {}

    # checking if it is already there in memo
    if targetSum in memo:
        return memo[targetSum]

    # checking for base cases
    if targetSum == 0:
        return []
    if targetSum < 0:
        return None

    for num in numbers:
        remainder = targetSum - num
        rem_res = howSum_memo(remainder, numbers, memo)

        if rem_res is not None:
            memo[targetSum] = rem_res + [num]
            return memo[targetSum]
    memo[targetSum] = None
    return None

if __name__ == "__main__":
    test_cases = [(7, [2, 3]), (7, [5, 3, 4, 7]), (300, [7, 14])]

    print("--- Testing Memoization Appraoch ---\n")
    for target, nums in test_cases:
```

```

result = howSum_memo(target, nums)
res = result if result is not None else "null"
print(f"howSum({target}, {nums}) -> {res}")

```

Output:

```

.../courses/spring_2026/infsci_2591_ad on
→ python hw3/memo_approach.py
--- Testing Memoization Approach ---

howSum(7, [2, 3]) -> [3, 2, 2]
howSum(7, [5, 3, 4, 7]) -> [4, 3]
howSum(300, [7, 14]) -> null

```

2. Tabulation approach code:

```

def howSum_tabu(targetSum, numbers):
    table = [None] * (targetSum + 1)
    table[0] = []

    for i in range(targetSum + 1):
        if table[i] is not None:
            for num in numbers:
                if i + num <= targetSum:
                    table[i + num] = table[i] + [num]
    return table[targetSum]

if __name__ == "__main__":
    test_cases = [(7, [2, 3]), (7, [5, 3, 4, 7]), (300, [7,
14])]
    print("--- Testing Tabulation ---\n")
    for target, nums in test_cases:
        result = howSum_tabu(target, nums)
        res = result if result is not None else "null"
        print(f"howSum({target}, {nums}) -> {res}")

```

Output:

```
.../courses/spring_2026/infsci_2591_ad on  
→ python hw3/tabulation_approach.py  
--- Testing Tabulation ---  
  
howSum(7, [2, 3]) -> [3, 2, 2]  
howSum(7, [5, 3, 4, 7]) -> [4, 3]  
howSum(300, [7, 14]) -> null
```